

Energy, power, delta functions and Fourier series

Energy, power and delta functions

Energy and power

- The total energy content of a signal $f(t)$ is defined as,

$$E_f = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

- The average power content of a signal $f(t)$ is defined as,

$$P_f = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

(this is the average over the time interval T).

* Usually only one of energy or power is defined.

Delta functions

- The delta function $\delta(t)$ can be defined in the following ways.

$$(i) \quad \delta(t) = \lim_{\varepsilon \rightarrow 0} f_1(t; \varepsilon), \quad f_1(t; \varepsilon) = \frac{1}{\varepsilon} \text{ for } t \in [-\frac{\varepsilon}{2}, \frac{\varepsilon}{2}], \text{ 0 o/w.}$$

$$(ii) \quad \delta(t) = \lim_{\varepsilon \rightarrow 0} f_2(t; \varepsilon), \quad f_2(t; \varepsilon) = \frac{\varepsilon}{\varepsilon^2 t^2 + \varepsilon}$$

$$(iii) \quad \delta(t) = \lim_{a \rightarrow 0} f_3(t; a), \quad f_3(t; a) = \frac{\sin(at)}{\pi t}$$

and have the following properties:

$$\delta(t) = 0 \text{ for } t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$\int_{-\infty}^{\infty} g(t) \delta(t - t_0) dt = g(t_0)$$

Sinc function

- The sinc function is defined as

$$\text{sinc}(x) = \frac{\sin x}{x}$$

- Its integral from $-\infty$ to ∞ is

$$\int_{-\infty}^{\infty} \text{sinc}(x) dx = \pi$$

consider the function $I(b) = \int_0^{\infty} \frac{\sin x}{x} e^{-bx} dx \rightarrow \int_{-\infty}^{\infty} \text{sinc}(x) dx = 2I(0)$.

$$\frac{dI}{db} = \int_0^{\infty} \frac{\partial}{\partial b} \frac{\sin x}{x} e^{-bx} dx$$

$$= - \int_0^{\infty} \sin x e^{-bx} dx$$

$$= - \frac{1}{b^2 + 1}$$

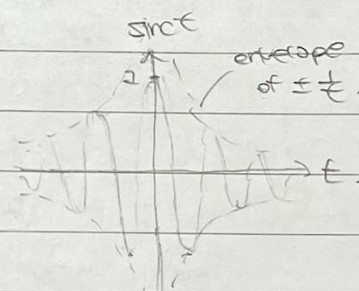
$$\therefore I(b) = - \int \frac{1}{b^2 + 1} db$$

$$= -\tan^{-1}(b) + C$$

As $b \rightarrow \infty$, $I(b) = \int_0^{\infty} \frac{\sin x}{x} e^{-bx} dx \rightarrow 0$, so $0 = -\frac{\pi}{2} + C \rightarrow C = \frac{\pi}{2}$.

$$\therefore I(b) = -\tan^{-1}(b) + \frac{\pi}{2}$$

$$\int_{-\infty}^{\infty} \text{sinc}(x) dx = 2[-\tan^{-1}(0) + \frac{\pi}{2}] = \pi$$



Reynold's technique of differentiating under the integral sign

A useful integral.

- Consider the integral I

$$\begin{aligned} I &= \int_{-\infty}^{\infty} \exp(j\omega t) d\omega = \lim_{A \rightarrow \infty} \int_{-A}^A \exp(j\omega t) d\omega \\ &= \lim_{A \rightarrow \infty} \left[\frac{e^{j\omega t}}{j t} \right]_{-A}^A \\ &= \lim_{A \rightarrow \infty} \left(2 \frac{\sin(At)}{t} \right) \\ &= \lim_{A \rightarrow \infty} 2\pi f_3(t; A) \end{aligned}$$

$$\therefore \int_{-\infty}^{\infty} \exp(j\omega t) d\omega = 2\pi \delta(t)$$

Fourier series.

Real and complex Fourier series.

- Any function $g(t)$, which is periodic in the interval $[t_0, t_0+T]$ has a real or complex Fourier series representation.

Real :

$$g(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos\left(\frac{2\pi n}{T}t\right) + b_n \sin\left(\frac{2\pi n}{T}t\right) \right\}$$

where $a_n = \frac{2}{T} \int_{t_0}^{t_0+T} g(t) \cos\left(\frac{2\pi n}{T}t\right) dt$, $b_n = \frac{2}{T} \int_{t_0}^{t_0+T} g(t) \sin\left(\frac{2\pi n}{T}t\right) dt$

complex :

$$g(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{T}t}$$

where $c_n = \frac{1}{T} \int_{t_0}^{t_0+T} g(t) e^{-j\frac{2\pi n}{T}t} dt$.

- By considering the expression for c_n in real (imaginary parts,

$$\begin{aligned} c_n &= \frac{1}{T} \int_{t_0}^{t_0+T} g(t) \cos\left(\frac{2\pi n}{T}t\right) dt - \frac{j}{T} \int_{t_0}^{t_0+T} g(t) \sin\left(\frac{2\pi n}{T}t\right) dt \\ &= \frac{1}{2} a_n - \frac{j}{2} b_n. \end{aligned}$$

$\therefore$ for $n > 0$, and real $g(t)$,

$$\begin{cases} 2c_n = a_n - j b_n \\ 2c_n^* = a_n + j b_n \end{cases}$$

$\Leftrightarrow$

$$\begin{cases} a_n = c_n^* + c_n \\ j b_n = c_n^* - c_n \end{cases}$$

- Also considering the expression for c_n ,

$$c_n^* = \left[\frac{1}{T} \int_{t_0}^{t_0+T} g(t) e^{-j\frac{2\pi n}{T}t} dt \right]^* = \frac{1}{T} \int_{t_0}^{t_0+T} g(t) e^{j\frac{2\pi n}{T}t} dt = \frac{1}{T} \int_{t_0}^{t_0+T} g(t) e^{-j\frac{2\pi(-n)}{T}t} dt = c_{-n}$$

i.e. $c_{-n} = c_n^*$

- the modulus of c_n is easily obtained in terms of the real coefficients.

$$2|c_n| = \sqrt{a_n^2 + b_n^2}$$

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Scaling, stretching and shifting.

- consider modifying $g(t)$ by scaling the amplitude by a factor of a , shifting it along the axis by b and changing the period to $T' = \beta T$. The modified signal $g_1(t)$ is.

$$g_1(t) = a g\left(\frac{t-b}{\beta}\right).$$

- Its Fourier series is hence

$$\begin{aligned} g_1(t) &= a \sum_{n=-\infty}^{\infty} c_n e^{j n \omega_0 \left(\frac{t-b}{\beta}\right)} \\ &= \sum_{n=-\infty}^{\infty} \left\{ a c_n e^{-j n \omega_0 \frac{b}{\beta}} \right\} e^{j n \omega_0' t} \\ &= \sum_{n=-\infty}^{\infty} c_n' e^{j n \omega_0' t} \end{aligned}$$

where the new coefficients and fundamental frequency are

$$c_n' = a c_n e^{-j n \omega_0 \frac{b}{\beta}}$$

;

$$\omega_0' = \frac{\omega_0}{\beta}$$

Differentiation and integration,

- suppose $g(t) = \sum_{n=-\infty}^{\infty} c_n e^{j n \omega_0 t}$, $h(t) = \frac{d}{dt} g(t)$, $f(t) = \int g(t) dt$.

$$(i) \quad h(t) = \frac{d}{dt} g(t) = \sum_{n=-\infty}^{\infty} \{j n \omega_0 c_n\} e^{j n \omega_0 t} = \sum_{n=-\infty}^{\infty} c_n' e^{j n \omega_0 t}$$

where

$$c_n' = j n \omega_0 c_n.$$

$$(ii) \quad f(t) = \int g(t) dt = C_0 t + \sum_{n \neq 0} \left\{ \frac{c_n}{j n \omega_0} \right\} e^{j n \omega_0 t} + K = C_0 t + \sum_{n \neq 0} c_n' e^{j n \omega_0 t} + K.$$

where

$$c_n' = \frac{1}{j n \omega_0} c_n$$

$$K = \frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt$$

* (ii) can only be a Fourier series if $C_0 = 0 \rightarrow f(t) = \sum_{n \neq 0} c_n' e^{j n \omega_0 t} + K.$

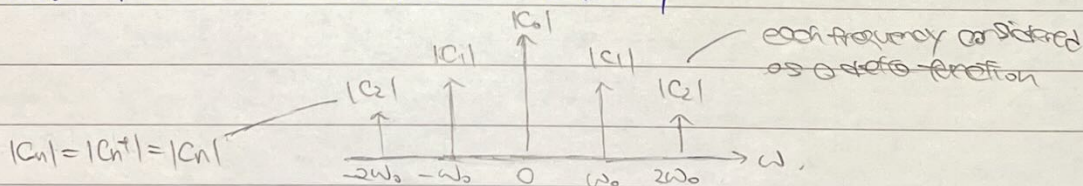
Interpretation of Fourier coefficients

- consider the complex Fourier series for a periodic signal $g(t)$.

$$g(t) = \sum_{n=-\infty}^{\infty} c_n e^{j n \omega_0 t}$$

where each complex exponential and its conjugate together can be regarded as a pure frequency component of the signal, w/ frequency $n \omega_0$.

- This frequency content can be represented graphically



- The amplitudes r_n of the harmonics are defined by rewriting the real Fourier series as follows.

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos(n \omega_0 t) + b_n \sin(n \omega_0 t)\} = r_0 + \sum_{n=1}^{\infty} r_n \sin(n \omega_0 t + \phi_n)$$

$$\text{so } r_0 = \frac{a_0}{2} \text{ and } r_n = \sqrt{a_n^2 + b_n^2}.$$

- since $2|c_n| = \sqrt{a_n^2 + b_n^2}$,

$$r_n = 2|c_n| = |c_n| + |c_{-n}|$$

for $n=0$,

$$r_0 = c_0.$$

Parseval's theorem for Fourier series.

- The magnitude of the complex coefficients can be related to the average power of the signal $g(t)$ over one period.

$$P_g = \frac{1}{T} \int_0^T |g(t)|^2 dt = \sum_{n=-\infty}^{\infty} |c_n|^2$$

→ This is Parseval's theorem for Fourier series.

The Fourier Transform

Mathematical formulation of the Fourier Transform

- Consider the complex Fourier series coefficient formula for the function $f(t)$

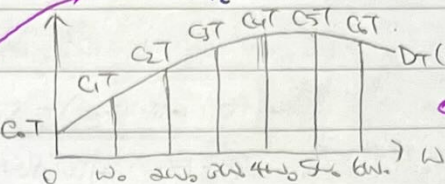
$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \quad \text{where } \omega_0 = \frac{2\pi}{T}$$

(Note that c_n is the Fourier series component w/ frequency $n\omega_0$)

- Now we can define a function of frequency which equals Tc_n at the pts $\omega = n\omega_0$, i.e.

$$D_T(n\omega_0) = Tc_n = \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \quad n \in \mathbb{Z}$$

the factor T normalises the integral, as the integral will be zero for most functions



the scale factor T doesn't affect the shape of the plot

Another function that satisfies such condition can be obtained by replacing $n\omega_0$ w/ a continuous variable ω ,

$$D_T(\omega) = \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt$$

- As we increase the period T , ω_0 decreases, and the discrete pts $D_T(n\omega_0)$ define the whole function $D_T(\omega)$ for all values of ω .

- The Fourier Transform $F(\omega)$ is obtained as the limit of $D_T(\omega)$ as $T \rightarrow \infty$, i.e.

$$F(\omega) = \lim_{T \rightarrow \infty} D_T(\omega) = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} f(t) e^{-j\omega t} dt$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

The Inverse Fourier Transform (IFT).

- It is possible to invert the Fourier Transform to determine uniquely the signal $f(t)$ from its Fourier transform $F(\omega)$.

- Starting w/ the forward Fourier Transform formula,

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Now multiply $F(\omega)$ by $e^{j\omega t}$ and integrate wrt ω .

$$I = \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t') e^{-j\omega t'} dt' e^{j\omega t} d\omega$$

$$= \int_{-\infty}^{\infty} f(t') \int_{-\infty}^{\infty} e^{j\omega(t-t')} d\omega dt'$$

[swap order of integration]

$$= \int_{-\infty}^{\infty} f(t') \cdot 2\pi \delta(t-t') dt'$$

$$\left[\int_{-\infty}^{\infty} e^{j\omega \tau} d\omega = 2\pi \delta(\tau) \right]$$

$$= 2\pi f(t)$$

[sifting theorem]

$$\therefore f(t) = \frac{1}{2\pi}$$

$\rightarrow$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

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Important Fourier Transforms.

① complex exponential.

$$f(t) = e^{j\omega_0 t}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-j(\omega - \omega_0)t} dt = 2\pi \delta(\omega - \omega_0)$$

* $\delta(\omega - \omega_0) = \delta(\omega_0 - \omega)$

$$\therefore F(\omega) = 2\pi \delta(\omega - \omega_0)$$

shifting prop.

② cosine

$$f(t) = \cos(\omega_0 t) = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$F(\omega) = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

③ sine

linearity of
Fourier transform

$$f(t) = \sin(\omega_0 t) = -\frac{j}{2}(e^{j\omega_0 t} - e^{-j\omega_0 t})$$

$$F(\omega) = j\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

④ Delta function

$$f(t) = a\delta(t)$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} a\delta(t) e^{-j\omega t} dt$$

$$\therefore F(\omega) = a$$

⑤ DC offset

$$f(t) = a$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = a \int_{-\infty}^{\infty} e^{-j\omega t} dt = a \cdot 2\pi \delta(\omega)$$

$$F(\omega) = 2\pi a \delta(\omega)$$

⑥ Rectangular pulse

$$f(t) = b \text{rect}\left(\frac{t}{T}\right)$$

$$\text{rect}(t) = \begin{cases} 1 & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0 & \text{o/w} \end{cases}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_{-\frac{T}{2}}^{\frac{T}{2}} b e^{-j\omega t} dt = -\frac{b}{j\omega} [e^{-j\omega t}]_{-\frac{T}{2}}^{\frac{T}{2}}$$

$$F(\omega) = bT \text{sinc}\left(\frac{\omega T}{2}\right)$$

⑦ Gaussian.

$$f(t) = e^{-a^2 t^2}$$

$$\frac{df(t)}{dt} = -2a^2 t e^{-a^2 t^2} = -2a^2 t f(t)$$

$$FT\{t f(t)\} = j \frac{dF(\omega)}{d\omega}$$

differentiation

$$j\omega F(\omega) = -2a^2 j \frac{dF(\omega)}{d\omega}$$

$$\int_0^{\omega} \omega d\omega = -2a^2 \int_0^{\omega} \frac{1}{F(\omega)} dF(\omega)$$

$$\frac{\omega^2}{2} = -2a^2 (\ln F(\omega) - \ln F(0))$$

$$F(\omega) = \frac{\sqrt{\pi}}{a} e^{-\frac{\omega^2}{4a^2}}$$

$$F(0) = \int_{-\infty}^{\infty} f(t) e^{j0t} dt = \int_{-\infty}^{\infty} f(t) dt = \frac{\sqrt{\pi}}{a}$$

the Fourier Transform of a
Gaussian shape is itself a Gaussian shape

Properties of the Fourier Transform

Linearity of Fourier Transforms

- consider the Fourier transform of the linear combination of $f(t)$ and $g(t)$.

$$\begin{aligned} \text{FT}[\alpha f(t) + \beta g(t)] &= \int_{-\infty}^{\infty} [\alpha f(t) + \beta g(t)] e^{-j\omega t} dt \\ &= \alpha \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt + \beta \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt \end{aligned}$$

$$\boxed{\text{FT}[\alpha f(t) + \beta g(t)] = \alpha \text{FT}[f(t)] + \beta \text{FT}[g(t)]}$$

i.e. the Fourier transform is a linear operation

Derivatives.

- Let the Fourier transform of $f(t)$ be $F(\omega)$, then $\frac{df}{dt}$ is given by

$$\frac{df}{dt} = \frac{d}{dt} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \right) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \frac{d}{dt} e^{j\omega t} d\omega.$$

$$\therefore \frac{df}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} j\omega F(\omega) e^{j\omega t} d\omega$$

$$\rightarrow \boxed{\text{FT}\left[\frac{df}{dt}\right] = j\omega F(\omega)}$$

- In general, the Fourier transform of $\frac{d^n f}{dt^n}$ is given by

$$\boxed{\text{FT}\left[\frac{d^n f}{dt^n}\right] = (j\omega)^n F(\omega)}$$

Multiplication by t .

- Consider the derivative of $F(\omega)$ wrt ω .

$$\begin{aligned} \frac{d}{d\omega} F(\omega) &= \frac{d}{d\omega} \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) \frac{d}{d\omega} e^{-j\omega t} dt \\ &= -j \int_{-\infty}^{\infty} t f(t) e^{-j\omega t} dt \\ &= -j \text{FT}[t f(t)] \end{aligned}$$

$$\therefore \boxed{\text{FT}[t f(t)] = j \frac{dF(\omega)}{d\omega}}$$

Similarity theorem (time scaling).

- Let the Fourier transform of $f(t)$ be $F(\omega)$, then

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

substitute αt for t ,

$$f(\alpha t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega \alpha t} d\omega.$$

For $\alpha > 0$, substitute $\omega' = \alpha \omega$, so $d\omega = \frac{1}{\alpha} d\omega'$

$$f(\alpha t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\alpha} F\left(\frac{\omega'}{\alpha}\right) e^{j\omega' t} d\omega'$$

$$\therefore \boxed{\text{FT}[f(\alpha t)] = \frac{1}{\alpha} F\left(\frac{\omega'}{\alpha}\right)}$$

also true for $\alpha < 0$.

i.e. if we stretch in the time domain, we contract in the frequency domain and vice versa.

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Helsenberg-Gabor principle.

- The Helsenberg-Gabor principle states if any function $f(t)$ has time duration T , and its Fourier Transform $F(\omega)$ has frequency bandwidth B , then.

time-bandwidth product. $\boxed{TB \geq 1}$

- This formalises the idea that stretching in the domain results in contracting in the frequency domain ($FT[f(at)] = \frac{1}{|a|} F(\frac{\omega}{a})$).

Frequency shift and modulation.

- Let the Fourier transform of $f(t)$ be $F(\omega)$, then

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

substitute $\omega - \omega_0$ for ω ,

$$F(\omega - \omega_0) = \int_{-\infty}^{\infty} f(t) e^{-j(\omega - \omega_0)t} dt = \int_{-\infty}^{\infty} e^{j\omega_0 t} f(t) e^{-j\omega t} dt$$

$$\therefore \boxed{FT[e^{j\omega_0 t} f(t)] = F(\omega - \omega_0)}$$

- since $f(t) \cos(\omega_0 t) = \frac{1}{2} [e^{j\omega_0 t} f(t) + e^{-j\omega_0 t} f(t)]$, by linearity,

$$\boxed{FT[\cos(\omega_0 t) f(t)] = \frac{1}{2} F(\omega - \omega_0) + \frac{1}{2} F(\omega + \omega_0)}$$

i.e. the spectrum of $f(t)$ is separated into two parts, each half of its original strength, and shifted along the ω axis by $\pm \omega_0$.

Time shift

- Let the Fourier transform of $f(t)$ be $F(\omega)$, then

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

Substitute $t - t_0$ for t ,

$$f(t - t_0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega(t - t_0)} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-j\omega t_0} F(\omega) e^{j\omega t} d\omega$$

$$\therefore \boxed{FT[f(t - t_0)] = e^{-j\omega t_0} F(\omega)}$$

Duality

- Let the Fourier transform of $f(t)$ be $g(\omega)$, then.

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} g(\omega) e^{j\omega t} d\omega$$

Substitute $-\omega'$ for ω , and rename ω w/ t' ,

$$f(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} g(t') e^{j\omega t'} dt'$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} g(t') e^{-j\omega t'} dt'$$

$$\therefore \boxed{FT[g(t)] = 2\pi f(\omega)}$$

i.e. if we have one Fourier transform pair

then we automatically have the dual Fourier Transform pair.

$$FT[f(t)] = g(\omega)$$

$$FT[g(t)] = 2\pi f(\omega)$$

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The multiplication theorem and Parseval's theorem

- consider two functions $f_1(t)$ and $f_2(t)$ w/ FTS $F_1(\omega)$ and $F_2(\omega)$. The integral of the product of f_1 and f_2^* is given by

$$\begin{aligned}\int_{-\infty}^{\infty} f_1(t) f_2^*(t) dt &= \int_{-\infty}^{\infty} f_1(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} F_2(\omega) e^{j\omega t} d\omega \right]^* dt \\ &= \int_{-\infty}^{\infty} f_1(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} F_2^*(\omega) e^{-j\omega t} d\omega dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F_2^*(\omega) \int_{-\infty}^{\infty} f_1(t) e^{-j\omega t} dt d\omega\end{aligned}$$

$$\boxed{\int_{-\infty}^{\infty} f_1(t) f_2^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F_1(\omega) F_2^*(\omega) d\omega} \quad [\text{Multiplication theorem}]$$

- putting $f_2(t) = f_1(t)$ in the above leads to Parseval's theorem.

$$\boxed{\int_{-\infty}^{\infty} |f_1(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F_1(\omega)|^2 d\omega} \quad [\text{Parseval's theorem}]$$

i.e. the energy of a signal can be found either by integrating in the time domain or in the frequency domain, and that energy in time domain $\propto$ energy in frequency domain.

- The energy of a time function $f(t)$ between frequencies ω_1 and ω_2 is given by

$$\boxed{E = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} |F(\omega)|^2 d\omega + \frac{1}{2\pi} \int_{-\omega_2}^{-\omega_1} |F(\omega)|^2 d\omega}$$

for real valued functions $f(t)$,

$$\boxed{E = 2 \cdot \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} |F(\omega)|^2 d\omega}$$

for real-valued $f(t)$,
 $F^*(\omega) = F(-\omega)$, so
 $|F(-\omega)|^2 = |F^*(\omega)|^2 = |F(\omega)|^2$

convolution and multiplication.

- Let $h(t)$ be the convolution of $f(t)$ and $g(t)$, and their FTS be $F(\omega)$, $G(\omega)$, $H(\omega)$.

$$h(t) = f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau$$

The Fourier transform of $h(t)$ is therefore.

$$H(\omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau e^{-j\omega t} dt$$

using $u = t - \tau$, $dt = du$,

$$\begin{aligned}H(\omega) &= \int_{-\infty}^{\infty} f(\tau) \int_{-\infty}^{\infty} g(u) e^{-j\omega(u+\tau)} du d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) e^{-j\omega\tau} d\tau \int_{-\infty}^{\infty} g(u) e^{-j\omega u} du \\ &= F(\omega) G(\omega)\end{aligned}$$

$$\boxed{\text{FT}[f(t) * g(t)] = \text{FT}[f(t)] \text{FT}[g(t)]}$$

- By duality, it can be shown that

$$\boxed{\text{FT}[f(t) \cdot g(t)] = \frac{1}{2\pi} \text{FT}[F(\omega)] * \text{FT}[G(\omega)]}$$

i.e. convolution in one domain is equivalent to multiplication in the other domain.

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Symmetry.

- Consider the conjugate of the Fourier Transform of a real function $f(t)$, $F^*(\omega)$

$$\begin{aligned} F^*(\omega) &= \left[\int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \right]^* \\ &= \int_{-\infty}^{\infty} f^*(t) e^{j\omega t} dt \\ &= \int_{-\infty}^{\infty} f(t) e^{j\omega t} dt. \end{aligned}$$

substituting $\rightarrow$ for ω in the definition of the forward Fourier Transform, we have.

$$F(-\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$\therefore \boxed{F(-\omega) = F^*(\omega)}$$

- It can also be shown that

$\rightarrow$ if $f(t)$ is real and even, then $F(\omega)$ is real and even

$\rightarrow$ if $f(t)$ is real and odd, then $F(\omega)$ is imaginary and odd.

Fourier Transform and Laplace Transform.

- The definitions for the Fourier and Laplace Transforms are as follows:

$$\rightarrow \text{Fourier: } F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$\rightarrow \text{Laplace: } \int_0^{\infty} f(t) e^{-st} dt$$

- the integrand is the same if we set $s = j\omega$, but the limits of integration are different.

- For a function $f(t)$ that is zero for $t < 0$, we have that

$$\boxed{F(\omega) = \tilde{f}(j\omega)}$$

provided that both integrals exist.

- For a causal LTI system, w/ impulse response $g(t)$, the frequency response is given by

$$\boxed{\text{Frequency response} = G(\omega) = \tilde{g}(j\omega)}$$

- DES can be solved using either Fourier Transforms or Laplace Transforms.

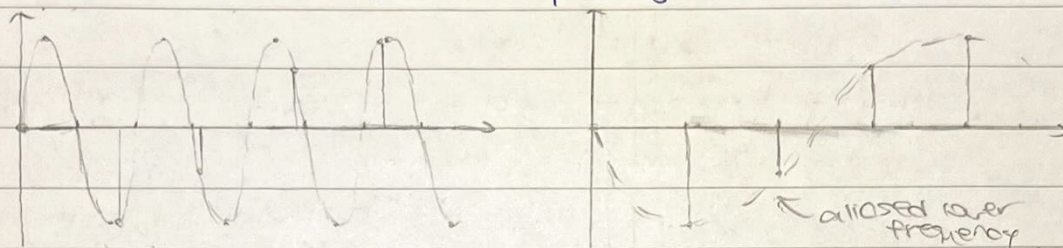
$\rightarrow$ Fourier is better suited to steady-state analysis

$\rightarrow$ Laplace is better suited to problems w/ BC of $t=0$.

Sampling theory

Digital sampling.

- Suppose a continuous time signal is given by $f(t)$, $-\infty < t < +\infty$.
choose a sampling interval T and read off the value of $f(t)$ at $t=nT$, $n \in \mathbb{Z}$.
The obtained values $f(nT)$ are the sampled version of $f(t)$.
- We need to choose the sampling interval T carefully, since
 - ↳ Too large a value of T will mean loss of detail from $f(t)$ (aliasing).
 - ↳ Too small a value means unnecessary storage of over-detailed / redundant data.



Mathematical representation of the sampled signal

- We can find a mathematical representation of the sampled signal using a train of delta functions.
- First, consider a single sample at $t=nT$. We multiply the sample $f(nT)$ and multiply it by a delta function centred at $t=nT$ to get

$$f(nT) \delta(t-nT).$$

since $\delta(t-nT) = 0$ except at $t=nT$, we can rewrite this as

$$f(t) \delta(t-nT).$$

The whole sampled signal is simply given by the sum of all such samples,

$$f_s(t) = \sum_{n=-\infty}^{\infty} f(t) \delta(t-nT)$$

$$= f(t) \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

$$f_s(t) = f(t) \delta_p(t)$$

train of impulses,
w/ period T .

As $\delta_p(t)$ is a periodic function, it can be represented as a Fourier series

$$\delta_p(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

$$\text{where } \omega_0 = 2\pi/T \text{ and } c_n = \frac{1}{T} \int_{-T/2}^{T/2} \delta_p(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jn\omega_0 t} dt = \frac{1}{T} \forall n.$$

sampling frequency.

$$\therefore f_s(t) = f(t) \sum_{n=-\infty}^{\infty} \frac{1}{T} e^{jn\omega_0 t}$$

$$f_s(t) = \frac{1}{T} \sum_{n=-\infty}^{\infty} f(t) e^{jn\omega_0 t}$$

- $f_s(t)$ is a continuous-time signal which contains only the sampled data information $f(nT)$, and is zero elsewhere.

* $f_s(t)$ is only a conceptual version of the sampled signal — in no way are we implying that there are infinite / infinite energy in a real sampled signal.

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Discrete-time Fourier Transform (DTFT).

- The Fourier Transform of the sampled signal $f_s(t)$, is the discrete-time Fourier Transform (DTFT), $F_s(\omega)$. It can be written as follows:

$$\begin{aligned} F_s(\omega) &= \int_{-\infty}^{\infty} f_s(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \left[\sum_{n=-\infty}^{\infty} f(t) \delta(t - nT) \right] e^{-j\omega t} dt \\ &= \sum_{n=-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) e^{-j\omega t} \delta(t - nT) dt \right] \\ F_s(\omega) &= \sum_{n=-\infty}^{\infty} f(nT) e^{-j\omega nT} \end{aligned}$$

- The DTFT gives the frequency content of the ideal sampled signal $f_s(t)$.

Nyquist frequency and reconstruction.

- Consider the ideal sampled signal $f_s(t) = \frac{1}{T} \sum_{n=-\infty}^{\infty} f(t) e^{jn\omega_0 t}$. The Fourier transform of each term in the summation is given by

$$FT[f(t) e^{jn\omega_0 t}] = F(\omega - n\omega_0) \quad [\text{frequency shift}]$$

where $F(\omega)$ is the FT of $f(t)$.

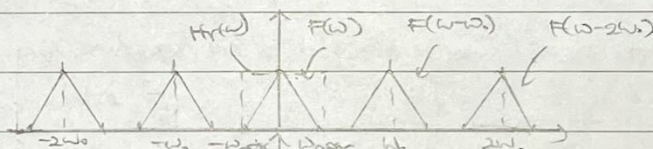
By linearity of the FT, the FT of the whole ideal sampled signal $f_s(t)$, $F_s(\omega)$ is given by

$$F_s(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_0)$$

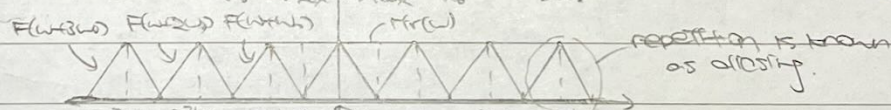
i.e. the FT of the idealised sampled signal is simply $\frac{1}{T} \times$ the FT of the continuous signal repeated every integer multiple of the sampling frequency and summed together

- consider the sampled spectrum $F_s(\omega)$ of a signal w/ BW ω_{max} and for various sampling frequency $\omega_0 = 2\pi/T$:

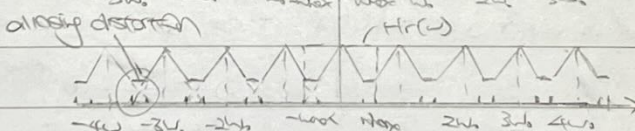
(a) $\omega_0 = 3\omega_{max}$



(b) $\omega_0 = 2\omega_{max}$



(c) $\omega_0 = 1.6\omega_{max}$



- For cases (a), (b), we can apply a filter w/ frequency response $H_r(\omega)$ to reconstruct the original spectrum $F(\omega)$ perfectly; for case (c), the original spectrum $F(\omega)$ is not properly constructed. (only possible when there is no overlap b/w the periodic repetitions of $F(\omega)$).

- The Nyquist sampling theorem states that for a signal $f(t)$ that has a max. frequency content (BW ω_{max}), it is possible to reconstruct $f(t)$ perfectly from its sampled version $f_s(t)$, provided the sampling frequency is at least $\omega_0 \geq 2\omega_{max}$.

- The min. sampling frequency is the Nyquist frequency ω_{Nyq} .

$$\omega_{Nyq} = 2\omega_{max}$$

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Ideal reconstruction filter.

- The ideal filter frequency response for perfect reconstruction is the rectangular pulse

$$H_r(\omega) = T \cdot \text{rect}\left(\frac{\omega}{2\omega_{\max}}\right)$$

- The impulse response of the filter is then the IFT of $H_r(\omega)$, which is

$$h_r(t) = \frac{\omega_{\max} T}{\pi} \text{sinc}(\omega_{\max} t)$$

- If we are sampling at exactly the Nyquist frequency ($\omega_{\max} = \frac{\omega_s}{2}$), then,

$$h_r(t) = \frac{\omega_s T}{2\pi} \text{sinc}\left(\frac{\omega_s t}{2}\right) = \text{sinc}\left(\frac{\omega_s t}{2}\right)$$

Since multiplication in the frequency domain is equivalent to convolution in the time domain,

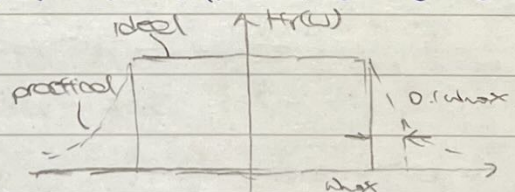
$$F(\omega) = H_r(\omega) F_s(\omega) \quad \Leftrightarrow \quad f(t) = h_r(t) * f_s(t)$$

$$\begin{aligned} \therefore f(t) &= \int_{-\infty}^{\infty} f_s(\tau) \text{sinc}\left(\frac{\omega_s(t-\tau)}{2}\right) d\tau \\ &= \int_{-\infty}^{\infty} \left[\sum_{n=-\infty}^{\infty} f(nT) \delta(\tau - nT) \right] \text{sinc}\left(\frac{\omega_s(t-\tau)}{2}\right) d\tau \\ &= \sum_{n=-\infty}^{\infty} f(nT) \int_{-\infty}^{\infty} \delta(\tau - nT) \text{sinc}\left(\frac{\omega_s(t-\tau)}{2}\right) d\tau \\ &= \sum_{n=-\infty}^{\infty} f(nT) \text{sinc}\left(\frac{\pi}{T}(t - nT)\right) \end{aligned}$$

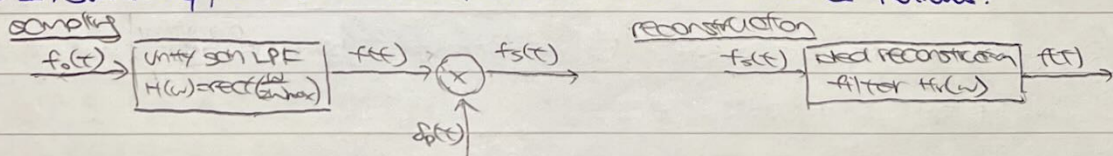
The above expression is an exact interpolation formula for determining $f(t)$ from its samples $f(nT)$.

Practical considerations.

- We must determine the max. frequency component $\omega_{\max}$ present in a signal before processing.
- To eliminate the aliasing effects of high frequency noise or unwanted high signal frequencies, we first filter the data w/ a unity gain LPF (frequency response $H(\omega) = \text{rect}\left(\frac{\omega}{2\omega_{\max}}\right)$).
- Then, perform sampling at the Nyquist rate $\omega_s = 2\omega_{\max}$ to get digital samples $f(nT)$.
- Reconstruct the signal by passing the sampled signal through the ideal reconstruction filter. (In practice, we cannot exactly implement the ideal reconstruction filter $H_r(\omega)$. — we must allow ~10% extra signal BW for the transition band of the filter.)



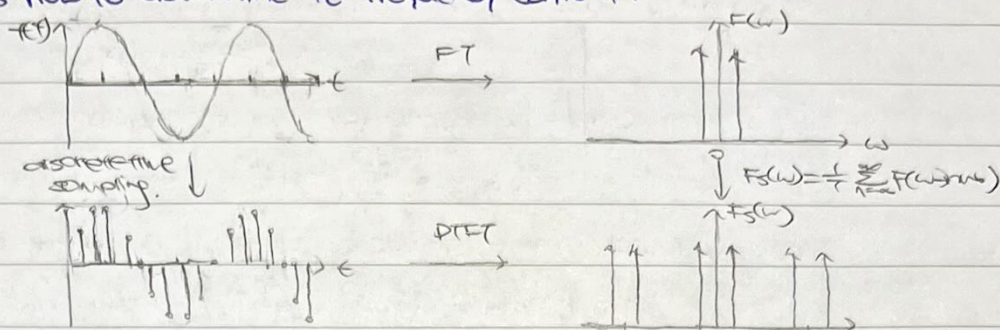
- In summary, a signal is sampled and reconstructed as follows:



The discrete Fourier transform

The discrete Fourier transform (DFT).

- Given the sampled values of a signal $f(t)$, i.e. $\{f(t=0), \dots, f(t=T), f(0), f(T), \dots, f(0)\}$, the DFT shows how to determine the frequency content.



- However in practice, there are two reasons why we can't compute the DTFT of a signal.

- ↳ 1) We don't have access to all of the data pts from $-\infty$ to ∞ (even if we did, the computing time would be infinite)
- ↳ 2) $F_s(\omega)$ is defined over the continuous range of frequencies from $-\infty$ to ∞ . $\rightarrow$ cannot calculate and store $F_s(\omega)$ for all frequency values [uncountably infinite]

- The discrete Fourier transform (DFT) can bypass such practical difficulties.

- ↳ 1) Only consider data pts within some finite range — say $t=0, T, 2T, \dots, (N-1)T$.

The DTFT is modified to.

$$F_s^N(\omega) = \sum_{n=0}^{N-1} f(nT) e^{j\omega nT}$$

we assume $f(nT) = 0$ outside the range $n=0, \dots, N-1$

provided we have enough of the original continuous signal between $t=0$ and $t=nT$, the truncated DTFT will be similar to the full DTFT

- ↳ 2) Calculate only over a finite span of frequencies. As $F_s(\omega)$ is periodic w/ period ω_0 , we don't need to calculate values beyond the Nyquist frequency.

Evaluating $F_s(\omega)$ at N evenly spaced frequency values $\omega=0, \frac{\omega_0}{N}, \frac{2\omega_0}{N}, \dots, \frac{(N-1)\omega_0}{N}$, we have.

$$\begin{aligned} F_s^N\left(\frac{n\omega_0}{N}\right) &= \sum_{k=0}^{N-1} f(kT) e^{-j\omega \frac{n\omega_0}{N} T} \\ &= \sum_{k=0}^{N-1} f(kT) e^{-j\frac{2\pi}{N} kn} \end{aligned}$$

- The above is the DFT. Defining $F_n = F_s^N\left(\frac{n\omega_0}{N}\right)$ and $f_n = f(nT)$, we have

$$F_n = \sum_{k=0}^{N-1} f_k e^{-j\frac{2\pi}{N} kn}$$

- We can interpret component F_k as the frequency content of the signal at frequency $k\frac{\omega_0}{N}$ or $k\omega_0/N$.

- The DFT is periodic, i.e.

$$F_k = F_{k+N}$$

- For real signals, using the defn for DFT, we can show that

$$F_k = F_k^*$$

$$(F_{-k} = \sum_{n=0}^{N-1} f_n e^{-j\frac{2\pi}{N} (-k)n} = \sum_{n=0}^{N-1} f_n e^{j\frac{2\pi}{N} kn} ; F_k^* = \sum_{n=0}^{N-1} f_n^* (e^{-j\frac{2\pi}{N} kn})^* = \sum_{n=0}^{N-1} f_n e^{j\frac{2\pi}{N} kn})$$

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Inverse discrete Fourier Transform (IDFT).

- Consider multiplying the forward DFT by $e^{jkm2\pi/N}$ and sum over $m=0$ to $N-1$.

$$\begin{aligned}\sum_{m=0}^{N-1} F_m e^{jkm2\pi/N} &= \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} f_n e^{-jnm2\pi/N} e^{jkm2\pi/N} \\ &= \sum_{n=0}^{N-1} f_n \sum_{m=0}^{N-1} e^{j(k-n)m2\pi/N}\end{aligned}$$

The inner summation over m is a GP w/ common ratio $r = e^{j(k-n)2\pi/N}$, so

$$\begin{aligned}\sum_{m=0}^{N-1} e^{j(k-n)m2\pi/N} &= \frac{1 - e^{j(k-n)2\pi}}{1 - e^{j(k-n)2\pi/N}} = \begin{cases} 0 & \text{if } k \neq n \\ N & \text{if } k = n \end{cases} \\ \therefore \sum_{m=0}^{N-1} F_m e^{jkm2\pi/N} &= \sum_{n=0}^{N-1} f_n \cdot 0 + f_k \cdot N\end{aligned}$$

- The expression for the inverse discrete Fourier Transform is therefore given by

$$f_k = \frac{1}{N} \sum_{m=0}^{N-1} F_m e^{jkm2\pi/N}.$$

other considerations

- Fast algorithms - fast method exist for converting the DFT, known generically as the Fast Fourier Transform (FFT).

- The effects of truncating a data sequence (when taking a finite set of samples) might introduce some discontinuities at the start and end of the data frame $\rightarrow$ high frequency components. $\rightarrow$ this can be avoided by applying a window to the sampled data.

Summary of various Fourier Transforms.

① Fourier Transform (FT) and Inverse Fourier Transform (IFT)

$$\text{FT: } F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$\text{IFT: } f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

② Discrete time Fourier Transform (DTFT)

$$\text{DTFT: } F_s(\omega) = \sum_{n=-\infty}^{\infty} f(nT) e^{-j\omega nT} = \frac{1}{T} \sum_{k=-\infty}^{\infty} F(\omega - n\omega_0)$$

③ Discrete Fourier Transform (DFT) and Inverse Discrete Fourier Transform (IDFT)

$$\text{DFT: } F_m = \sum_{n=0}^{N-1} f_n e^{-jnm2\pi/N} \quad [0 \leq m \leq N-1] \quad \text{IDFT: } f_n = \frac{1}{N} \sum_{m=0}^{N-1} F_m e^{jnm2\pi/N} \quad [0 \leq n \leq N-1]$$